Designing an LLM-Based Application for Collaborative Learning

Ayoub EL KHAIARI
Project repository available at: GitHub

1 Learning Scenario

1.1 Educational Context

The proposed system is designed for a higher-education setting in the domain of Deep Learning, specifically within courses that introduce students to `PyTorch`. The system is intended for undergraduate or early graduate students in computer science, artificial intelligence, or data science programs.

Students at this level typically possess foundational knowledge in Python programming and linear algebra, but are transitioning toward more advanced topics such as neural network architectures, transfer learning, data preprocessing pipelines, and training optimization strategies. In this phase, learners often struggle not only with implementation but also with forming coherent conceptual models of how different components interact within a deep learning workflow.

1.2 Learners and Learning Objectives

The primary learners targeted by this system aim to understand the role of `torchvision` in image-based deep learning tasks and how it integrates within the broader PyTorch ecosystem. They seek to interpret and correctly apply data transformations such as `ToTensor` and normalization, particularly in relation to model training requirements. They are also expected to use pre-trained models and implement transfer learning strategies effectively, while reasoning about decisions such as freezing layers and selecting which components to fine-tune.

The objective is not simply to obtain correct answers, but to develop conceptual understanding through structured dialogue. Learners are expected to explain their reasoning, justify assumptions, compare alternatives, and refine their thinking through iterative interaction.

1.3 Relevance of LLM Collaboration

Collaboration with a Large Language Model (LLM) is particularly relevant in this context because deep learning concepts require iterative reasoning rather than one-step solutions. Students benefit from guided questioning, scaffolding, and structured reflection instead of direct answer delivery.

In this system, the LLM is not positioned as an authoritative solution provider. Instead, it functions as a conversational partner that supports active reasoning. Through multi-turn dialogue, the system encourages learners to articulate their ideas, confront misconceptions, and progressively refine their understanding.

2 View of Collaboration

Collaborative learning in this project is understood as a process in which knowledge is co-constructed through dialogue, shared problem solving, explanation, justification, and negotiation of meaning [1]. Learning is not treated as passive reception of information but as an interactive construction process.

The LLM adopts a collaborative stance that differs from default answer-generation behavior. It asks questions before answering, encourages learners to explain their reasoning, invites revision and reflection, and builds upon learner contributions. Rather than asserting truth, it negotiates meaning. Learning therefore occurs during interaction itself.

2.1 Shared Goal Orientation

A central principle is the explicit establishment of a shared learning goal. Both learner and LLM are oriented toward understanding and reasoning improvement rather than answer correctness. The LLM periodically restates or stabilizes this goal to prevent conceptual drift.

For example, instead of delivering an explanation, the system may ask: *"Let us understand why this approach works. What do you think is the key assumption here?"* This reframes the exchange as joint inquiry.

2.2 Mutual Contribution and Dialogue Structure

Collaboration requires active contribution from both participants. The learner must explain, justify, compare, and reflect. The LLM adapts to the learner's reasoning and avoids compensating for disengagement by providing complete answers prematurely. Interaction is structured as multi-turn dialogue rather than single exchange. A typical pattern follows: question, partial insight, learner reasoning, refinement, and deeper questioning. Each concept unfolds over multiple turns, enabling progressive clarification.

2.3 Scaffolding and Productive Friction

The system integrates scaffolding strategies that guide reasoning without removing cognitive effort, aligning with socio-constructivist perspectives in which guided interaction supports learners within their Zone of Proximal Development [3].

Prompts such as "What assumption are you making?" or *"Can you think of a counterexample?"* direct attention toward conceptual structure.

Additionally, the system introduces productive friction—mild epistemic challenge that stimulates deeper reasoning. For instance, the LLM may respond: *"I'm not fully convinced yet. Can you explain why this step follows?"* Such moments of tension enhance conceptual robustness.

2.4 Role of the LLM

The LLM acts as facilitator and co-explorer rather than instructor. It stabilizes shared goals, anchors discussion around key concepts, introduces partial insights, asks reflective questions, and maintains conceptual continuity. Full explanations are intentionally delayed to preserve learner cognitive effort.

3 Reflection and Analysis

3.1 Design Choices

3.1.1 Choice of Learning Scenario

Deep learning education was selected because it inherently requires reasoning across mathematical, architectural, and implementation levels. Students often struggle when transitioning from theory to applied frameworks such as PyTorch, making this domain suitable for studying collaborative AI support. This design choice also responds to prior findings that human–AI interaction often defaults to answer acceptance rather than genuine collaborative engagement, motivating the need for structured dialogical behaviors [2].

3.1.2 Three Experimental Approaches

Retrieval-Augmented Generation (RAG) The initial approach implemented RAG with collaborative prompting. Although dialogical behaviors were present, retrieval was frequently too broad and introduced cross-domain drift. Shared goals became unstable, resulting in interactive yet incoherent dialogue.

A refined version improved goal stabilization but did not resolve retrieval misalignment. This revealed three failure layers: prompt functioning correctly, retrieval lacking topic dominance, and learner experiencing cognitive overload (Confusion). The experiment demonstrated that collaborative prompting cannot compensate for poor retrieval alignment.

Prompt Engineering Only The second configuration removed retrieval and relied exclusively on structured collaborative prompting. This produced stable shared goals, strong learner activation, consistent multi-turn reasoning, and productive friction. Conceptual coherence improved significantly; however, external grounding in course material was lost.

Hybrid Approach The hybrid system combined retrieval with explicit collaborative constraints. Retrieved documents were treated as shared artifacts rather than direct answers, references were labeled, and domain-consistency constraints were added.

While structure improved, retrieval occasionally disrupted reasoning trajectories and misconceptions were not always surfaced explicitly. This exposed a core tension between maintaining collaboration and ensuring conceptual precision.

3.2 Collaboration Quality

Collaboration quality differed across the three configurations and revealed clear trade-offs.
The prompt-only approach produced the strongest dialogical structure. Goal stabilization, multi-turn reasoning, and productive friction worked effectively, resulting in coherent and cognitively engaging dialogue. However, the system lacked grounding in authoritative course material, limiting epistemic anchoring.

In contrast, the RAG-only configuration introduced retrieval grounding but frequently caused cross-domain drift. Semantically similar yet pedagogically misaligned passages were retrieved, destabilizing the shared goal. Although the interaction remained collaborative in form, it often revolved around incoherent or fragmented content. This resulted in what can be described as *collaborative nonsense* rather than collaborative learning.

The hybrid approach improved stability by combining structured prompting with constrained retrieval. It preserved dialogical coherence while introducing grounding, but collaboration quality remained sensitive to retrieval precision. Small domain misalignments could still disrupt reasoning trajectories.
Overall, effective collaborative learning requires both strong dialogical control and tightly aligned retrieval.

3.3 Limitations

Several system-level limitations were identified.

First, the retrieval pipeline relies primarily on semantic similarity without enforcing topic dominance constraints. As a result, passages from adjacent but distinct conceptual clusters may be retrieved simultaneously. This produces incoherent anchoring and fragmented dialogue progression.

Second, the system lacks an explicit misconception detection mechanism. When learners propose incorrect causal explanations (e.g., linking normalization to hardware acceleration), the collaborative stance preserves dialogue structure but does not automatically detect conceptual conflict.

Third, dialogue state is not formally tracked. Although the prompt enforces turn-level structure, there is no structured representation of evolving learner hypotheses, assumptions, or resolved subgoals. This limits longitudinal coherence across multiple turns.

These findings indicate that dialogical structure alone is insufficient. Pedagogical coherence requires tighter integration between retrieval modeling, dialogue state representation, and misconception detection.

3.4 Future Improvements

Future work will focus on integrating structured retrieval control and cognitive modeling mechanisms.

First, topic-dominance filtering can be introduced using clustering over embedding space (e.g., k-means or hierarchical clustering) to ensure that retrieved passages belong to the same conceptual cluster before being presented. Alternatively, a reranking stage using cross-encoders (e.g., MiniLM cross-encoder reranker) could prioritize pedagogically coherent passages over purely semantically similar ones.
Second, domain-consistency validation can be implemented using metadata tagging of course materials (task type, model family, objective function, architectural class). Retrieval would then enforce constraint-based filtering before similarity scoring.
Third, misconception detection could leverage contradiction classification models or natural language inference (NLI) frameworks to detect inconsistency between learner statements and retrieved reference material. This would allow the system to surface conflicts explicitly while maintaining collaborative tone.
Fourth, dialogue state tracking could be formalized through structured memory representations, such as: - Explicit hypothesis tracking, - Assumption logs, - Concept dependency graphs.
This would allow the system to monitor reasoning progression and detect conceptual drift dynamically.
Evaluation will combine interaction metrics (e.g., average reasoning turns per concept, goal restatement frequency) with cognitive indicators (e.g., conceptual coherence scoring, misconception correction latency, learner-reported cognitive load via standardized instruments such as NASA-TLX).
Together, these improvements aim to transform the system from prompt-level collaboration toward a retrieval-aware, state-aware pedagogical dialogue architecture.

4 Conclusion

This work demonstrates that collaborative LLM systems require tight integration between pedagogical design and retrieval control. Prompt-engineered collaboration can structure reasoning effectively, but retrieval alignment is essential for conceptual coherence.
Hybrid systems must balance epistemic grounding, dialogical continuity, and misconception handling. Effective collaborative AI therefore depends not only on behavioral prompting but also on structured retrieval design and dialogue-level consistency mechanisms.

References

[1] Pierre Dillenbourg. Chapter 1 (introduction) what do you mean by 'collaborative learning'? *Collaborative-learning: Cognitive and Computational Approaches*, Vol. 1, 01 1999.

[2] Catalina Gomez, Sue Min Cho, Shichang Ke, Chien-Ming Huang, and Mathias Unberath. Human-ai collaboration is not very collaborative yet: a taxonomy of interaction patterns in ai-assisted decision making from a systematic review. *Frontiers in Computer Science*, 6, jan 2025.

[3] Lev S. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press, 1978.